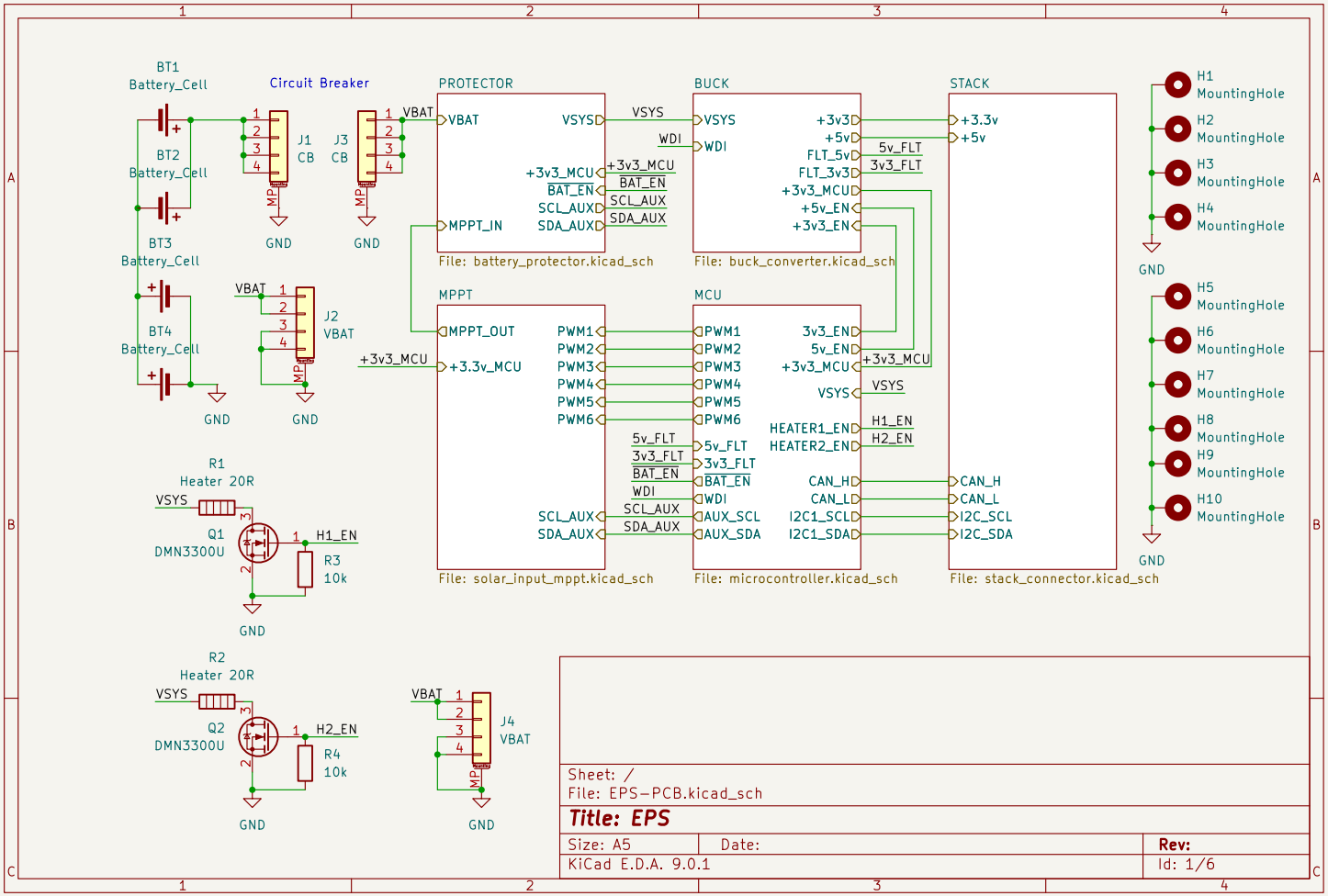
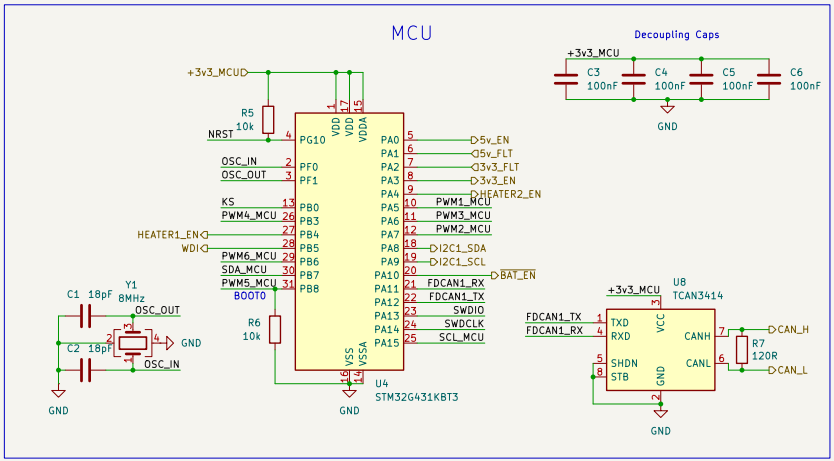
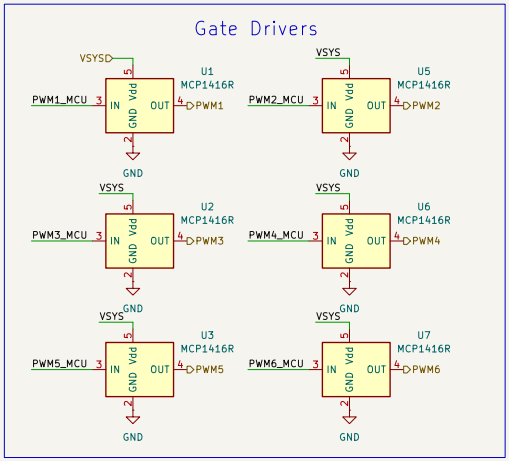
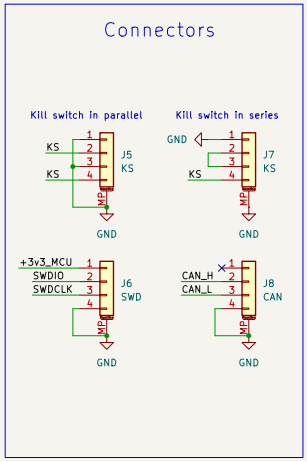

BT1
Battery_Cell
BT2
Battery_Cell
BT3
Battery_Cell
BT4
Battery_Cell
GND
Circuit Breaker
J1
CB
J3
CB
MP
GND
VBAT
J2
VBAT
PROTECTOR
VBAT
VSYS
MPPT_IN
+3v3_MCU
BAT_EN
SCL_AUX
SDA_AUX
File: battery_protector.kicad_sch
BUCK
VSYS
WDI
+3v3
+5v
FLT_5v
FLT_3v3
+3v3_MCU
+5v_EN
+3v3_EN
5v_FLT
3v3_FLT
File: buck_converter.kicad_sch
STACK
+3.3v
+5v
CAN_H
CAN_L
I2C_SCL
I2C_SDA
File: stack_connector.kicad_sch
MPPT
MPPT_OUT
+3.3v_MCU
PWM1
PWM2
PWM3
PWM4
PWM5
PWM6
5v_FLT
3v3_FLT
BAT_EN
WDI
SCL_AUX
SDA_AUX
File: solar_input_mppt.kicad_sch
MCU
PWM1
PWM2
PWM3
PWM4
PWM5
PWM6
5v_FLT
3v3_FLT
BAT_EN
WDI
AUX_SCL
AUX_SDA
3v3_EN
5v_EN
+3v3_MCU
VSYS
HEATER1_EN
HEATER2_EN
CAN_H
CAN_L
I2C1_SCL
I2C1_SDA
H1_EN
H2_EN
File: microcontroller.kicad_sch
R1
Heater 20R
Q1
DMN3300U
R3
10k
H1_EN
R2
Heater 20R
Q2
DMN3300U
R4
10k
H2_EN
J4
VBAT
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
H5
MountingHole
H6
MountingHole
H7
MountingHole
H8
MountingHole
H9
MountingHole
H10
MountingHole
Sheet: /
File: EPS-PCB.kicad_sch
Title: EPS
Size: A5
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 1/6





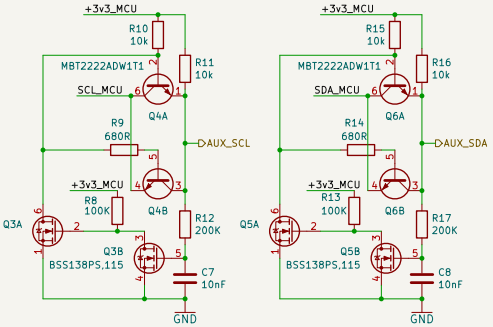


NOTE

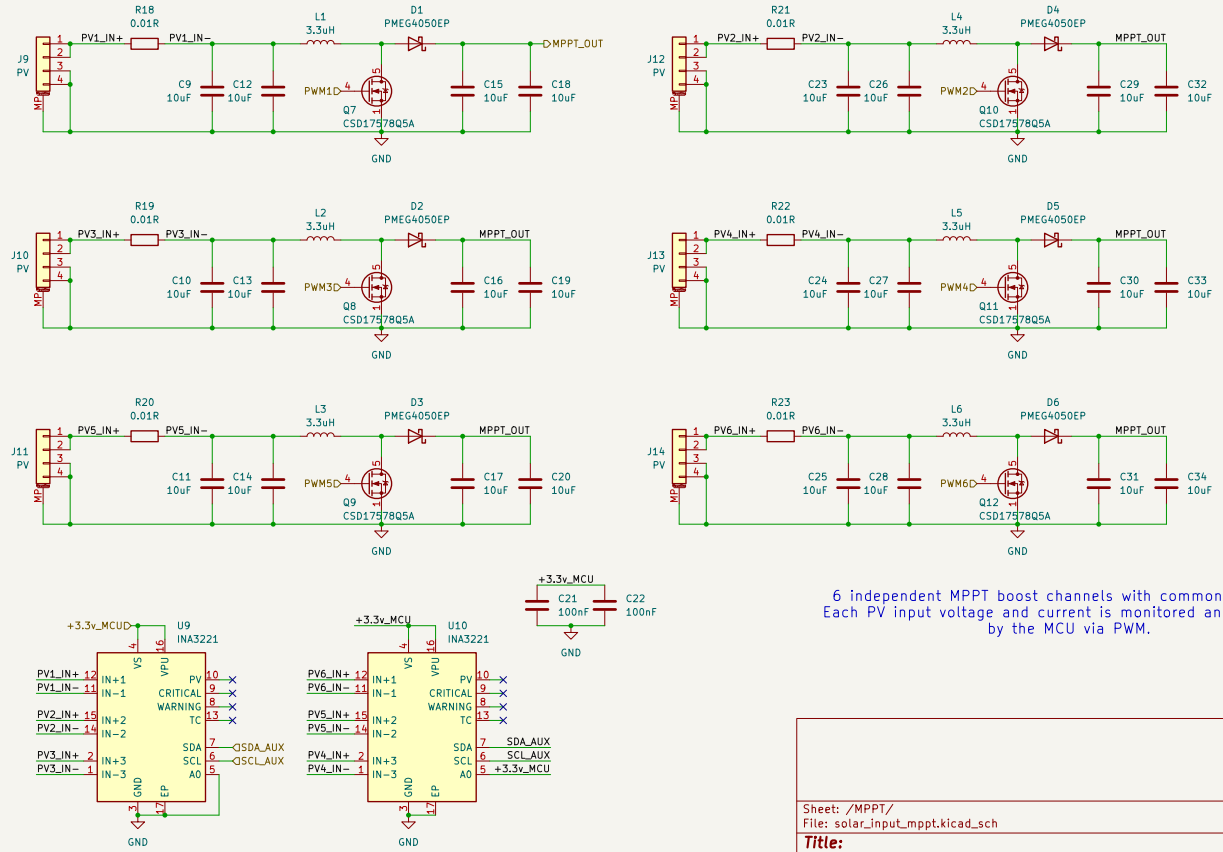
These novel bus protection circuits from PyCubed prevent traditional I2C/SPI failure modes where a single slave failure can disable the entire bus.

Learn more:
https://doi.org/10.36227/techrxiv.15166620

By default, slave clock and/or data lines can be held low and the Master (SAMD51) will still be able to communicate with the remainder of the bus.



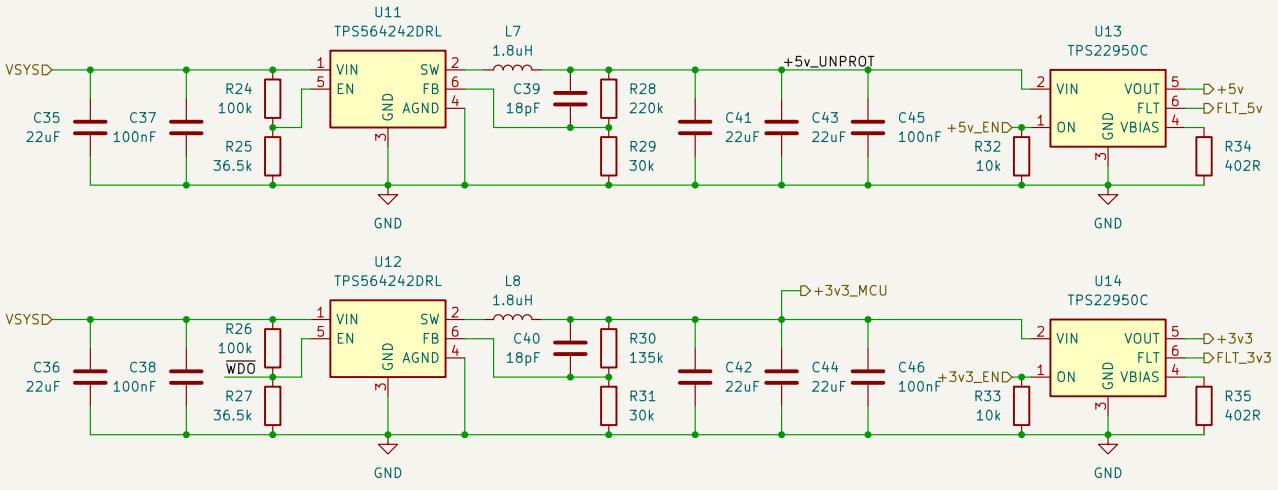
Sheet: /MCU/		
File: microcontroller.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 2/6



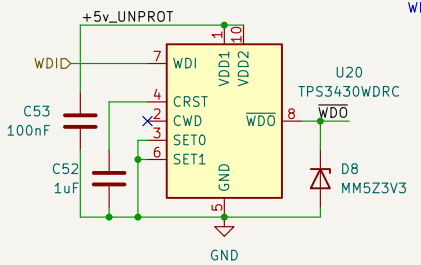
Sheet: /MPPT/
File: solar_input_mppt.kicad_sch

Title:

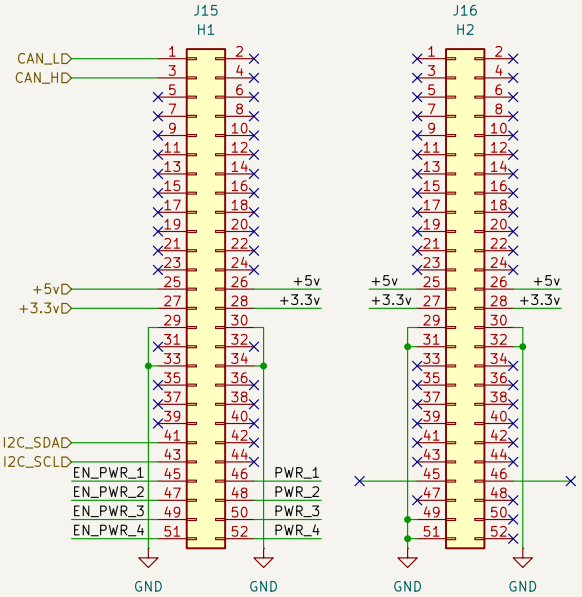
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 3/6



External Watchdog with window from 25.9ms to 46.8ms.
MCU should supply signal at 30 Hz with 50% duty.
When MCU failed to reset timer, WDO will be pulled down
and disable 3.3v converter for 3.2s.
Watchdog is powered by 5v to prevent cutoff.
Watchdog is disabled for 3.2s during power up.



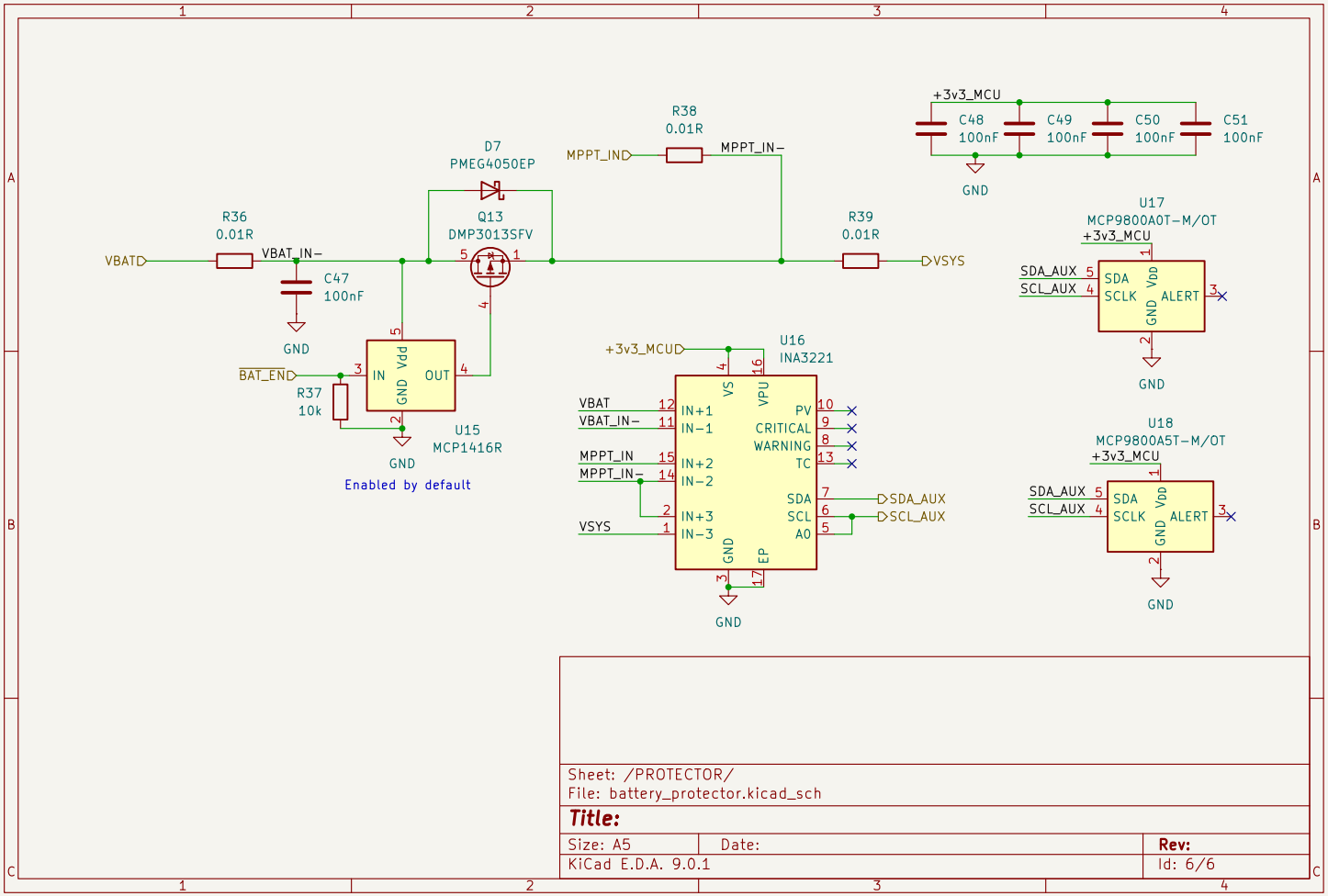
Sheet: /BUCK/		
File: buck_converter.kicad_sch		
Title:		
Size: A5	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 4/6



Sheet: /STACK/
File: stack_connector.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 5/6


+3v3_MCU
C48
100nF
C49
100nF
C50
100nF
C51
100nF
GND
R38
0.01R
MPPT_IN
MPPT_IN−
D7
PMEG4050EP
Q13
DMP3013SFV
R36
0.01R
VBAT
VBAT_IN−
C47
100nF
GND
R39
0.01R
VSYS
U17
MCP9800A0T−M/OT
+3v3_MCU
SDA_AUX
SCL_AUX
SDA
SCLK
Vdd
ALERT
GND
BAT_EN
R37
10k
IN
GND
Vdd
OUT
U15
MCP1416R
GND
Enabled by default
+3v3_MCU
U16
INA3221
VBAT
VBAT_IN−
MPPT_IN
MPPT_IN−
VSYS
IN+1
IN−1
IN+2
IN−2
IN+3
IN−3
VS
VPU
GND
EP
PV
CRITICAL
WARNING
TC
SDA
SCL
A0
SDA_AUX
SCL_AUX
GND
U18
MCP9800A5T−M/OT
+3v3_MCU
SDA_AUX
SCL_AUX
SDA
SCLK
Vdd
ALERT
GND
GND
Sheet: /PROTECTOR/
File: battery_protector.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 6/6